

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:21-12-23

Max. Marks: 300

Name of the Student: _____

H.T. NO:

21-12-23_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-1_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2025

PHYSICS

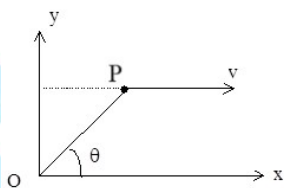
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

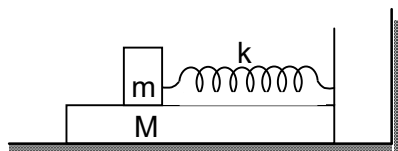
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A particle is moving parallel to x-axis as shown in the figure such that at all instants the y - component of its position vector is constant and is equal to 'b'. The angular velocity of the particle 'P' about the origin at the given instant is



- (A) $\frac{v}{b} \cos \theta$ (B) $\frac{v}{b} \sin \theta$ (C) $\frac{v}{b} \sin^2 \theta$ (D) vb
2. An object is placed on the surface of a smooth inclined plane of inclination θ . It takes time t to reach the bottom. If the same object is allowed to slide down a rough inclined plane of same inclination θ , so as to move the same distance, it takes time ' nt ' to reach the bottom where n is a number greater than 1. The coefficient of friction μ is given by:

- (A) $\mu = \tan \theta \left(1 - \frac{1}{n^2} \right)$ (B) $\mu = \cot \theta \left(1 - \frac{1}{n^2} \right)$
- (C) $\mu = \tan \theta \left(1 - \frac{1}{n^2} \right)^{\frac{1}{2}}$ (D) $\mu = \cot \theta \left(1 - \frac{1}{n^2} \right)^{\frac{1}{2}}$
3. A system of two bodies of masses ' m ' and ' M ' being interconnected by a spring of stiffness k , in its natural length, moves towards a rigid wall on a smooth horizontal surface as shown in figure with a K.E. of system ' E '. If the body M sticks to the wall after the collision, the maximum compression of the spring will be:

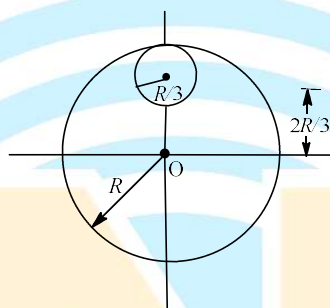


- (A) $\sqrt{\frac{mE}{Mk}}$ (B) $\sqrt{\frac{2mE}{(M+m)k}}$ (C) $\sqrt{\frac{2(m+M)E}{mk}}$ (D) $\sqrt{\frac{2ME}{(M+m)k}}$

4. A sphere of mass $m_1 = 2\text{kg}$ collides with a sphere of mass $m_2 = 3\text{kg}$ which is at rest. Mass m_1 , after the collision will move at right angle to the line joining centers of two spheres at the time of collision, assuming colliding surfaces are smooth, if the coefficient of restitution is

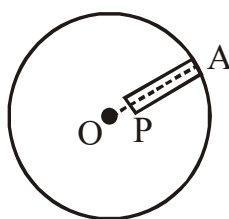
(A) $\frac{4}{9}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\sqrt{\frac{2}{3}}$

5. From a circular disc of radius R and mass $9M$, a small disc of radius $R/3$ is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is



A) $4MR^2$ B) $\frac{40}{9}MR^2$ C) $10MR^2$ D) $\frac{37}{9}MR^2$

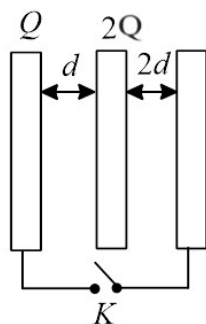
6. A planet of radius R has an acceleration due to gravity of g_s on its surface. A deep smooth tunnel is dug on this planet, radially inward, to reach a point P located at a distance of $\frac{R}{2}$ from the centre of the planet. Assume that the planet has uniform density. The kinetic energy required to be given to a small body of mass m , projected radially outward from P , so that it gains a maximum altitude equal to the thrice the radius of the planet from its surface, is equal to



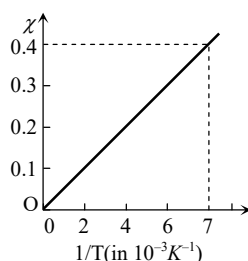
A) $\frac{63}{16}mg_sR$ B) $\frac{3}{8}mg_sR$ C) $\frac{9}{8}mg_sR$ D) $\frac{21}{8}mg_sR$

7. Water rises to a height of 10cm in a capillary tube and mercury falls to a depth of 3.42 cm in the same capillary tube. If the density of mercury is 13.6g/cc and the angle of contact of mercury and water are 135° and 0° respectively, the ratio of surface tension of water to mercury is
- (A) 1 : 0.15 (B) 1 : 3 (C) 1 : 6.5 (D) 1.5 : 1
8. Two moles of ideal helium gas are in a rubber balloon at 30°C . The balloon is fully expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is slowly changed to 35°C . The amount of heat required in raising the temperature is nearly (take $R=8.31 \text{ J/mol.K}$)
- A) 62 J B) 104 J C) 125 J D) 208 J
9. Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature T_0 , while box B contains one mole of helium at temperature $(7/3)T_0$. The boxes are then put into thermal contact with each other and heat flows between them until the gases reach a common final temperature (Ignore the heat capacity of boxes and heat exchange will happen only between boxes). Then, the final temperature of the gases, T_f , in terms of T_0 is
- A) $T_f = \frac{7}{3} T_0$ B) $T_f = \frac{3}{2} T_0$ C) $T_f = \frac{5}{2} T_0$ D) $T_f = \frac{3}{7} T_0$
10. A particle free to move along the x -axis has potential energy given by $U(x) = k[1 - \exp(-x^2)]$ for $-\infty \leq x \leq +\infty$, where k is a positive constant of appropriate dimensions. Then
- A) At point away from the origin, the particle is in unstable equilibrium
B) For any finite non-zero value of x , there is a force directed away from the origin
C) If its total mechanical energy is $k/2$, then its kinetic energy at the origin is k .
D) For small displacements from $x = 0$, the motion is simple harmonic.
11. The electric potential in a medium of dielectric constant 'unity' is $\phi(x, y, z) = ax^2$ where 'a' is a constant of suitable dimensions. The total charge contained in a cube of dimensions $-L \leq x, y, z \leq L$ is
- A) zero B) $-2a\epsilon_0 L^3$ C) $-16a\epsilon_0 L^3$ D) $-4a\epsilon_0 L^3$

12. Three large plates are arranged as shown. How much charge will flow through the key when it is closed?

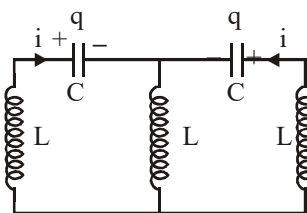


- A) $\frac{5Q}{6}$ B) $\frac{4Q}{3}$ C) $\frac{3Q}{2}$ D) Q
13. Assertion : A current I flows along the length of an infinitely long straight and thin walled pipe. Then, the magnetic field at any point inside the pipe is zero.
- Reason : $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$
- Read the assertion and reason carefully to mark the correct option out of the options given below:
- A) Both Assertion and Reason are true and the reason is the correct explanation of the assertion.
- B) Both Assertion and Reason are true but reason is not the correct explanation of the assertion.
- C) Assertion is true but Reason is false.
- D) Assertion and Reason both are false.
14. The $\chi - 1/T$ graph for an alloy of paramagnetic nature is shown in fig. the curie constant is nearly.

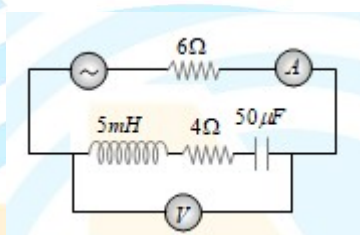


- A) **57 K** B) **$2.8 \times 10^{-3} K$** C) **570 K** D) **$17.5 \times 10^{-3} K$**

15. The natural frequency of the circuit shown in the figure is (Assume that equal currents i , i exist in the two branches, and the charges on the capacitors are equal)



- A) $\frac{1}{2\pi} \frac{1}{\sqrt{LC}}$ B) $\frac{1}{2\pi} \frac{1}{\sqrt{3LC}}$ C) $\frac{1}{\pi} \frac{1}{\sqrt{LC}}$ D) $\frac{1}{\pi} \sqrt{\frac{4}{3LC}}$
16. In the circuit shown in the figure, the ac source gives a voltage $V = 20 \cos(2000t)$. Neglecting source resistance, the voltmeter and ammeter reading will be ($\sqrt{2} \approx 1.4$)



- A) 0V, 0.47A B) 1.68V, 0.47A C) 0V, 1.4 A D) 5.6V, 1.4 A
17. A plane Electromagnetic Waves travelling along the positive X -direction has a wavelength of 3 mm. The variation in the electric field occurs in the Y -direction with an amplitude 66 Vm^{-1} . The equations for the electric and magnetic fields as a function of x and t are respectively
- A) $E_y = 66 \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right), \quad B_x = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- B) $E_y = 11 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right), \quad B_y = 11 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- C) $E_x = 66 \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right), \quad B_x = 2.2 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- D) $E_y = 66 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right), \quad B_x = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$
18. In young's double slit experiment, the distance between the slits varies with time as $d(t) = (2d_0 + d_0 \sin wt)$, where d_0 and 'w' are positive constants. The difference between the largest and the smallest fringe width obtained over time is ____

(D = distance between slits and screen $\gg d$ & λ = wavelength of light used)

- A) $\frac{D\lambda}{2d_0}$ B) $\frac{D\lambda}{3d_0}$ C) $\frac{2D\lambda}{3d_0}$ D) $\frac{D\lambda}{6d_0}$

19. Match the following.

List – I

A) Moderator

B) Control rods

C) Radiation shielding

D) Coolant

A) A – h, B – g, C – f, D – e

C) A – h, B – g, C – e, D – f

List – II

e) Absorbs heat

f) Prevents radiation from getting exposed to the outside

g) Absorb neutrons

h) Slow down neutrons

B) A – g, B – h, C – f, D – e

D) A – h, B – f, C – e, D – g

20. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of $\pm 0.05 \text{ mm}$ at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of $\pm 0.01 \text{ mm}$.

Take $g = 10.0 \text{ ms}^{-2}$ (exact). The Young's modulus obtained from the reading is

A) $(2.0 \pm 0.3) \times 10^{11} \text{ Nm}^{-2}$

B) $(2.0 \pm 0.2) \times 10^{11} \text{ Nm}^{-2}$

C) $(2.0 \pm 0.1) \times 10^{11} \text{ Nm}^{-2}$

D) $(2.0 \pm 0.05) \times 10^{11} \text{ Nm}^{-2}$

SECTION-II

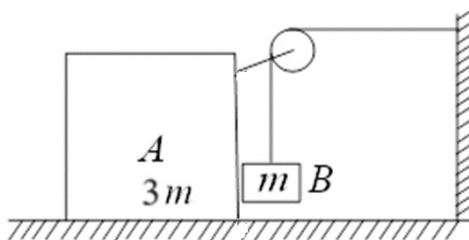
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

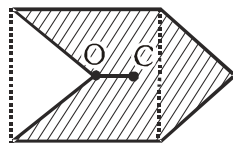
Marking scheme: +4 for correct answer, -1 in all other cases.

21. In the system shown all the surfaces are frictionless while pulley and string are massless.

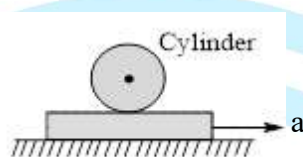
Mass of block A is $3m$ and that of block B is m . If the acceleration of block B with respect to ground after system is released from rest is 'a', then the value of '10 a' is (take $g=10 \text{ m/s}^2$ and $5\sqrt{2} = 7.0$)



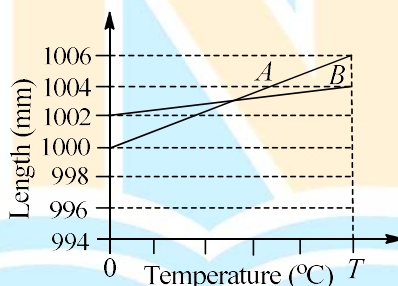
22. An arrow sign is made by cutting and re-joining a quarter part of a square plate of side 'L=1m' as shown. The distance OC, where 'C' is the centre of mass of the arrow, is(in cm)



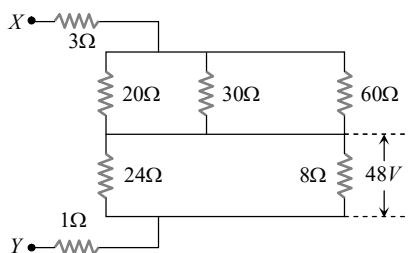
23. A uniform solid cylinder of mass M and radius R is placed on a rough horizontal board of same mass, which in turn is placed on a smooth surface. The coefficient of friction between the board and the cylinder is $\mu=0.3$. If the board starts accelerating with constant acceleration a , as shown in the figure, then the maximum value of a , so that the cylinder performs pure rolling is.....(in m/s^2) given $g=10\text{m/s}^2$



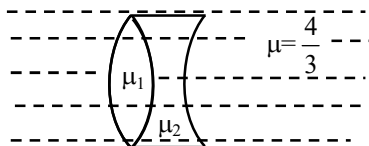
24. The variation of lengths of two metal rods A and B with change in temperature is shown in Figure. If the coefficients of linear expansion for the metal A is $\alpha_A = n \times 10^{-6} / ^\circ\text{C}$, then the value of 'n' will be, in nearest integer, given $\alpha_B = 9 \times 10^{-6} / ^\circ\text{C}$



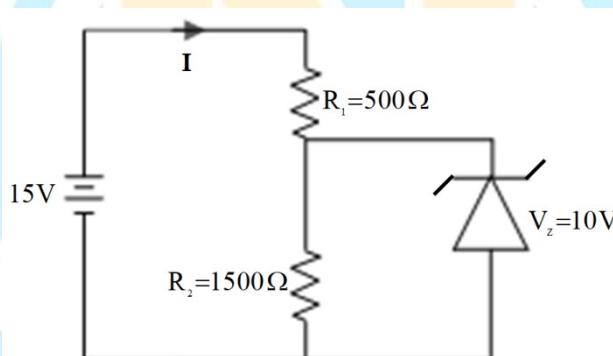
25. Two strings X and Y of a sitar produce a beat frequency of 4Hz . When the tension of the string Y is slightly increased the beat frequency is found to be 2Hz . If the frequency of X is 300Hz then the original frequency of Y was (in Hz)
26. The potential difference across 8 ohm resistance is 48 volt as shown in the figure. The value of potential difference across X and Y points will be.....(in Volts)



27. Two thin symmetrical lens of different nature have equal radii of curvature of all faces $R = 20$ cm. The lenses are put close together and immersed in water. The focal length of the system is 24 cm. The difference between refractive indices of the two lenses is $\times \frac{1}{9}$. Refractive index of water is $\frac{4}{3}$.



28. When a monochromatic point source of light is at a distance 0.2 m from a photoelectric cell, the saturation current and cut-off voltage are 12.0 mA and 0.5 V. If the same source is placed at 0.4 m away from the same photoelectric cell, then the saturation current, now, will be(in mA)
29. In the circuit given if the current through the Zener diode is I_z . Find the value of $6I_z$ (in mA)



30. In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of 11 k Ω are used. The figure of merit of the galvanometer is 60 μ A/division. In the absence of shunt resistance, the galvanometer produces a deflection of $\theta=9$ divisions when current flows in the circuit. The value of the shunt resistance (in Ω) that can cause the deflection of $\theta/2$, is closest to _____

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. If λ_0 is the threshold wavelength for photoelectric emission from a metal surface, λ is the wavelength of light falling on the surface of metal and m is the mass of electron, then the maximum speed of ejected electrons is given by

(A) $\left[\frac{2h}{m} (\lambda_0 - \lambda) \right]^{\frac{1}{2}}$

(B) $\left[\frac{2hc}{m} (\lambda_0 - \lambda) \right]^{\frac{1}{2}}$

(C) $\left[\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda} \right) \right]^{\frac{1}{2}}$

(D) $\left[\frac{2h}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right) \right]^{\frac{1}{2}}$

32. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

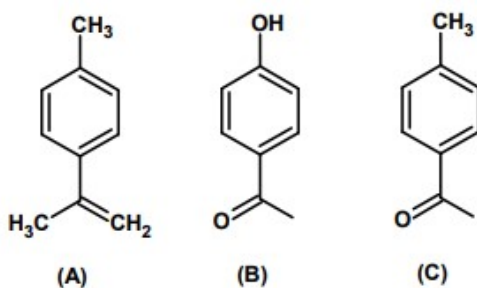
Assertion A: In TlI_3 which is isomorphous to CsI_3 , the metal is present in +1 oxidation state.

Reason R: Tl metal has fourteen f electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both A and R are correct and R is the correct explanation of A
- (B) A is not correct but R is correct
- (C) Both A and R are correct R is NOT the correct explanation of A
- (D) A is correct but R is not correct
33. Assertion (A): Bond dissociation energy of F_2 is smaller than that of Cl_2
- Reason (R): Bond dissociation energy is in order $F_2 < Cl_2 < Br_2 < I_2$
- A) Assertion is True, Reason is True; Reason is correct explanation for Assertion
- B) Assertion is True, Reason is True; Reason is NOT a correct explanation for Assertion
- C) Assertion is True, Reason is False
- D) Assertion is False, Reason is True

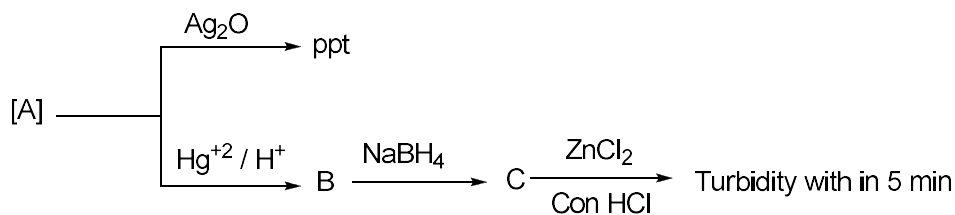
34. 5 g of urea is dissolved in one kg of water. If the solution is cooled to -0.200°C , then how many grams of ice would separate? (K_f of water = 1.86)
- (A) 200 (B) 225 (C) 325 (D) 175
35. A brown precipitate insoluble in excess of aqueous ammonia solution is formed by
- (A) $\text{FeCl}_3 + \text{NH}_4\text{Cl} \rightarrow$ (B) $\text{FeCl}_3 + \text{NH}_4\text{OH} \rightarrow$
(C) $\text{CrCl}_3 + \text{NH}_4\text{OH} \rightarrow$ (D) $\text{Hg}_2\text{Cl}_2 + \text{NH}_4\text{OH} \rightarrow$
36. For the first-order reaction $\text{A}_{(g)} \rightarrow 2\text{B}_{(g)} + \text{C}_{(g)}$, the total pressure after time t from the start of reaction with A is P and after infinite time, it is P_{∞} . Then the rate constant of the reaction is
- (A) $\frac{1}{t} \ln \frac{P_{\infty}}{P}$ (B) $\frac{1}{t} \ln \frac{2P_{\infty}}{3(P_{\infty} - P)}$
(C) $\frac{1}{t} \ln \frac{2P_{\infty}}{3P_{\infty} - P}$ (D) $\frac{1}{t} \ln \frac{2P_{\infty}}{P_{\infty} - 3P}$
37. The order of $E^{\circ}_{\text{M}^{3+}/\text{M}^{2+}}$ value is
- (A) $\text{Co} > \text{Mn} > \text{Fe}$ (B) $\text{Mn} > \text{Co} > \text{Fe}$ (C) $\text{Co} > \text{Fe} > \text{Mn}$ (D) $\text{Mn} > \text{Fe} > \text{Co}$
38. A chromatography column packed with silica gel (used as stationary phase) is used for the separation of



The column was then eluted with a mixture of Dichloro methane-nitromethane in 80:20 ratio, the sequence of elution of A, B and C is

- A) A, B, C (B) C, B, A (C) B, C, A (D) A, C, B

39. Consider the following reactions, What is A



- A) Acetylene B) propyne C) ethylene D) 2-butyne

40. In azomethane, H_3CNNCH_3 , what are the molecular geometries around the carbon and nitrogen atoms, respectively?

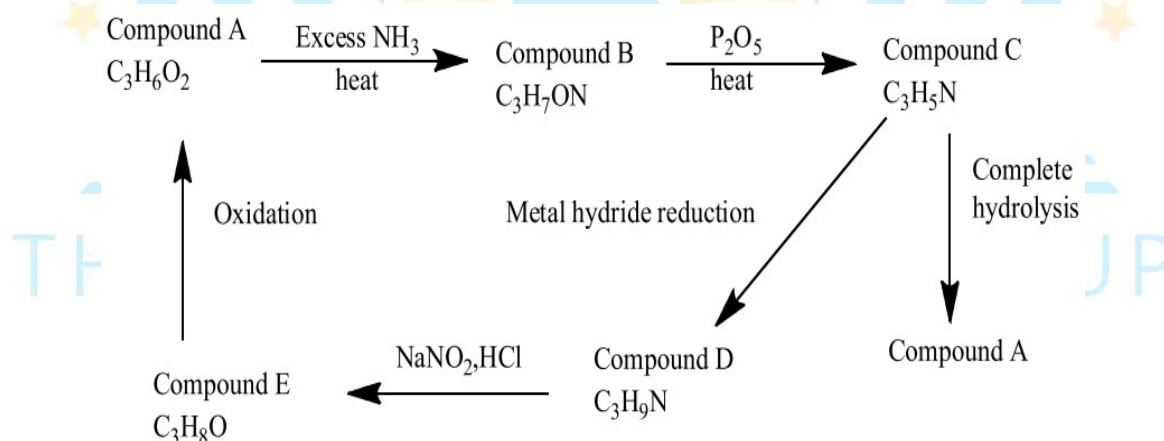
- A) Tetrahedral at carbon, bent at nitrogen
 B) Tetrahedral at carbon, linear at nitrogen
 C) Square planar at carbon, bent at nitrogen
 D) Square planar at carbon, linear at nitrogen

41. An organic compound contains only carbon, hydrogen, nitrogen, and oxygen. It is 61.71% C, 4.03% H, and 16.00% N by mass. What is its empirical formula?

- A) $\text{C}_5\text{H}_4\text{NO}$ B) $\text{C}_9\text{H}_7\text{N}_2\text{O}_2$ C) $\text{C}_{10}\text{H}_8\text{N}_2\text{O}$ D) $\text{C}_{11}\text{H}_8\text{NO}_2$

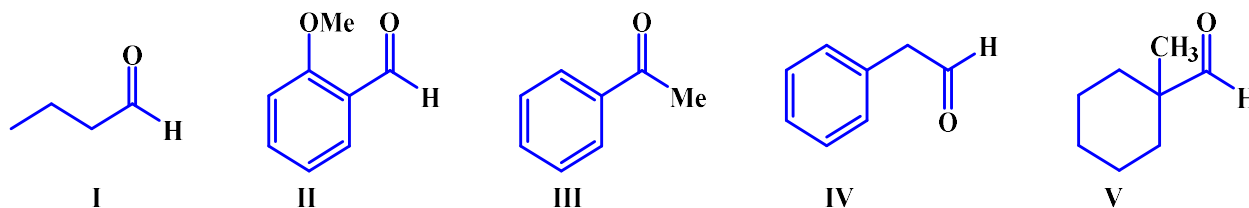
42. Consider the following sequence of reactions:

The functional groups undergoing change in the conversion of A to E respectively are



- A) $-\text{COOH}, -\text{NC}, -\text{CONH}_2, -\text{NH}_2, -\text{CH}_2\text{OH}$
 B) $-\text{COOH}, -\text{CN}, -\text{CONH}_2, -\text{NH}_2, -\text{CH}_2\text{OH}$
 C) $-\text{COOH}, -\text{CONH}_2, -\text{CN}, -\text{CH}_2\text{NH}_2, -\text{CH}_2\text{OH}$
 D) $-\text{CONH}_2, -\text{COOR}, -\text{NC}, -\text{NHR}, >\text{CHOH}$

43.



The compounds which can be reduced with formaldehyde and conc. AqKOH, are

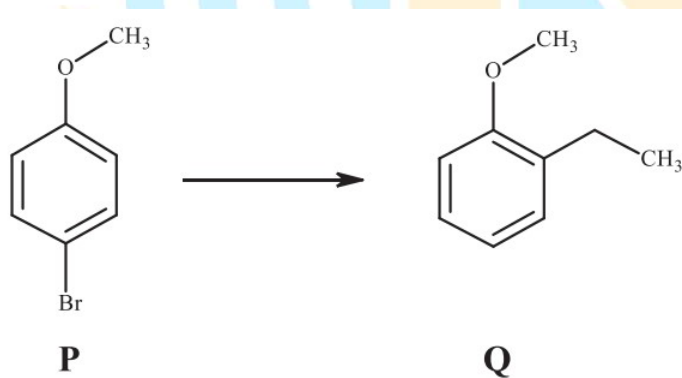
- A) only II and V B) only I and V
C) only II and III D) only I, II and IV

44. What is the X for given reaction?



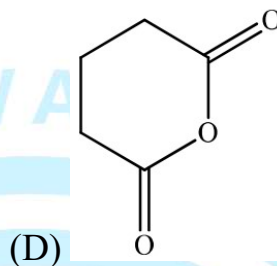
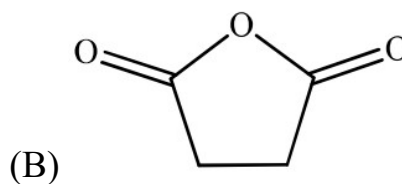
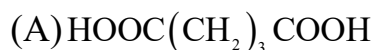
- A) B) C) D) All

45. The correct sequence of reactions to get 'Q' as the only product from 'P' is



- A) (i) H_2 & Pt catalyst (ii) $\text{C}_2\text{H}_5\text{Cl}$ & AlCl_3
B) (i) Mg in ether (ii) aqueous alcohol (iii) $\text{C}_2\text{H}_5\text{Cl}$ & AlCl_3
C) (i) Mg in ether (ii) $\text{C}_2\text{H}_5\text{Cl}$ & AlCl_3
D) (i) $\text{C}_2\text{H}_5\text{Cl}$ & AlCl_3 (ii) Mg in ether (iii) aqueous alcohol

46. When acid 'X' is heated to 230°C , along with CO_2 and , a compound 'Y' is formed. If 'X' is $\text{HOOC}(\text{CH}_2)_2\text{CH}(\text{COOH})_2$, the structure of 'Y' is



47. The best reagents and conditions to accomplish the following conversion is?



- A) (i) LiAlH_4 in ether, (ii) 3 moles of CH_3I followed by heating with AgOH
 B) (i) LiAlH_4 in ether; (ii) P_2O_5 and heat
 C) (i) 20 % H_2SO_4 & heat, (ii) P_2O_5 and heat
 D) H_2 and Lindlar catalyst
48. Two students did a set of experiments on ketones 'X' and 'Y' independently and obtained the following results.

Reaction /Experiment	X	Y
Optical rotation	Yes	Yes
Optical rotation after treatment with a base	Zero	Yes
NH_2NH_2 , KOH, Heat	Formation of an optically inactive hydrocarbon C_6H_{12}	Formation of an optically inactive hydrocarbon C_6H_{12}

The ketones 'X' and 'Y' are respectively

- (A) 2-ethylcyclobutanone and 3-ethylcyclobutanone
- (B) 2-methylcyclopentanone and 3-methylcyclopentanone
- (C) 3-methylcyclopentanone and 2-methylcyclopentanone
- (D) 3-methyl-4-penten-2-one and 4-methyl-1-penten-3-one

49. RNA and DNA are chiral molecules, their chirality is due to:

- A) D-sugar compound B) L-sugar component
- C) chiral bases D) Chiral phosphate ester groups

50. The electronic configurations of bivalent europium and trivalent cerium are (atomic number : Xe = 54, Ce = 58, Eu = 63)

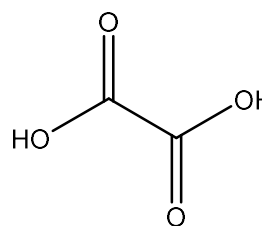
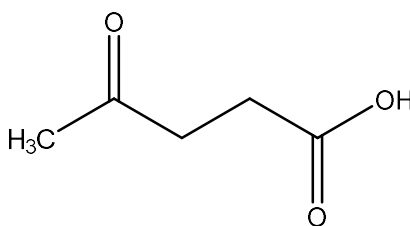
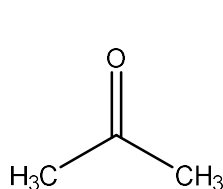
- (A) $[\text{Xe}]4f^2$ and $[\text{Xe}]4f^7$ (B) $[\text{Xe}]4f^7$ and $[\text{Xe}]4f^1$
- (C) $[\text{Xe}]4f^7 6s^2$ and $[\text{Xe}]4f^2 6s^2$ (D) $[\text{Xe}]4f^4$ and $[\text{Xe}]4f^9$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

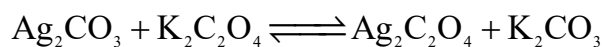
Marking scheme: +4 for correct answer, -1 in all other cases.

51. Geraniol is an acyclic unsaturated alcohol $\text{C}_{10}\text{H}_{18}\text{O}$, a terpene found in rose oil, adds two moles of bromine to form a tetrabromide, $\text{C}_{10}\text{H}_{18}\text{OBr}_4$. It can be oxidized to a ten-carbon aldehyde or to a ten-carbon carboxylic acid. Upon vigorous oxidation, geraniol yields:



Number of double bonds in Geraniol is?

52. The solubility product of $\text{Ag}_2\text{C}_2\text{O}_4$ at 25°C is $2.3 \times 10^{-11} \text{ M}^2$. A solution of $\text{K}_2\text{C}_2\text{O}_4$ containing 0.15 moles in 500 ml water is shaken at 25°C with Ag_2CO_3 till the following equilibrium is reached.



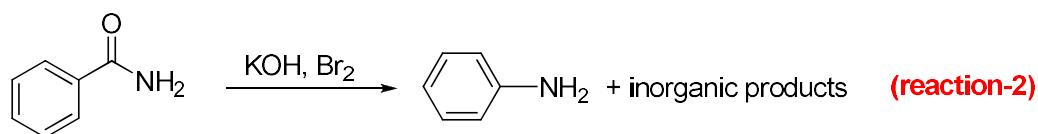
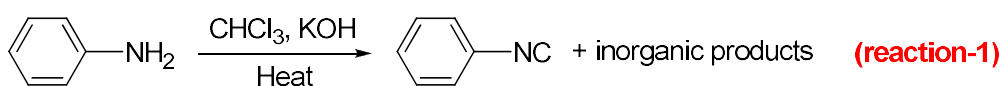
At equilibrium, the solution contains 0.035 mole of K_2CO_3 . Assuming the degree of dissociation of $\text{K}_2\text{C}_2\text{O}_4$ and K_2CO_3 to be equal, The solubility product of Ag_2CO_3 is $x \times 10^{-12}$, then x is _____

53. The volume, in mL, of 0.02 M $\text{K}_2\text{Cr}_2\text{O}_7$ solution required to react with 0.288 g of ferrous oxalate in acidic medium is _____. (Molar mass of Fe = 56 g mol⁻¹)
54. Equivalence conductance at infinite dilution of NH_4Cl , NaOH and NaCl are 129.8, 217.4 and 108.9 $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, respectively. If the equivalent conductance of 0.01 N solution of NH_4OH is 9.532 $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, then the percentage (%) dissociation of NH_4OH at this temperature is _____
55. Glycine ($\text{C}_2\text{H}_5\text{O}_2\text{N}$) is the simplest of amino acids. Molecular formula of the linear oligomer synthesized by linking ten glycine molecules together via a condensation reaction is $\text{C}_w\text{H}_x\text{O}_y\text{N}_z$

What is the value of $w+x+y+z$?

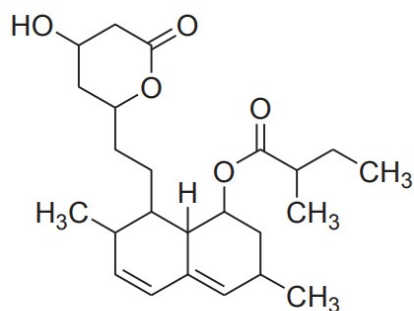
56. For silver, $C_p (\text{JK}^{-1} \text{ mol}^{-1}) = 23 + 0.01T$. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be close to _____

57.



If X is the moles of KOH consumed per mole of reactant in reaction-1 and Y is the moles of KOH consumed per mole of reactant in reaction 2, what is (X+Y)

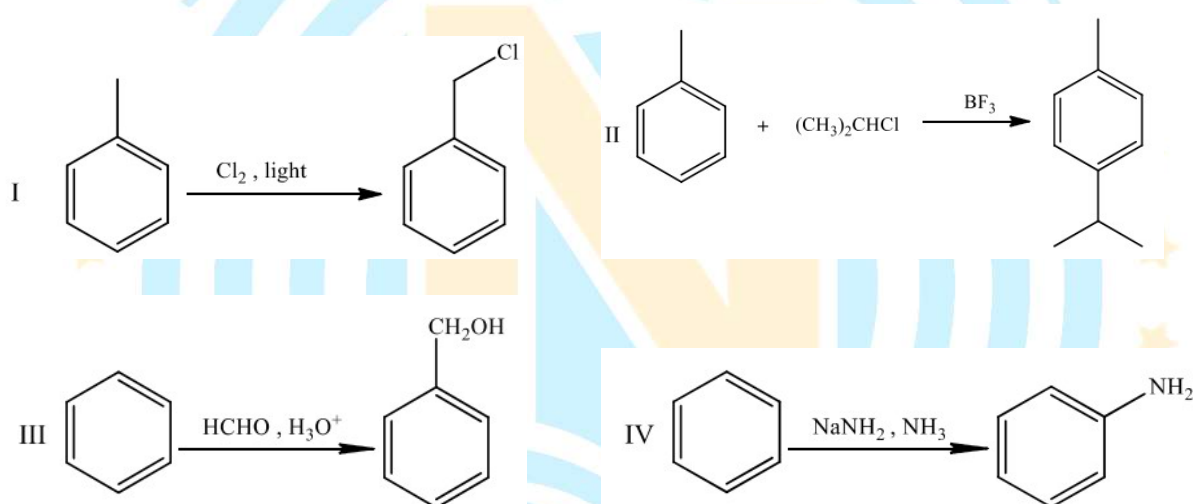
58. Lovastatin, a drug used to reduce the risk of cardio vascular diseases has the following structure



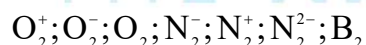
Lovastatin

The number of stereogenic centers present in lovastatin is

59. How many of the following are an example of aromatic electrophilic substitution reactions



60. Total number of paramagnetic species in which unpaired electron(s) is/are present in π^* M.O.



MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. The number of points in the complex plane that satisfy the conditions $|z - 2| = 2$,

$$z(1-i) + \bar{z}(1+i) = 4 \text{ (where } i = \sqrt{-1} \text{) is}$$

- A) 0 B) 1 C) 2 D) more than two

62. Let A and B are square matrices of same order satisfying $AB = A$ and $BA = B$, then

$$(A^{2019} + B^{2019})^{2020} \text{ is equal to:}$$

- A) $A + B$ B) $2020(A + B)$ C) $2^{2019}(A + B)$ D) $2^{2020}(A + B)$

63. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x^2 + 1 & 2-x & x-3 \\ x^2 - 3 & x+4 & 3x \end{vmatrix}$ then p is equal to

- A) -5 B) -4 C) -3 D) -2.

64. If α is the root of the equation $x^2 - x + 2 = 0$ then the value of $\frac{6(-\alpha^3 + 2\alpha^2 - \alpha)}{\alpha^5 - 3\alpha^4 + 3\alpha^3 - \alpha^2}$ is equal to:

- A) 3 B) 6 C) 9 D) 12

65. Number of 4 digit numbers of the form $N = abcd$ which satisfy following three conditions:

- i) $4000 \leq N < 6000$ ii) N is a multiple of 5

iii) $3 \leq b < c \leq 6$ is equal to:

- A) 12 B) 18 C) 24 D) 48

66. If a and b are chosen randomly by throwing a pair of fair cubical dice, then the

probability that $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{2}{x}} = 6$ equals:

- A) $\frac{4}{9}$ B) $\frac{2}{9}$ C) $\frac{3}{9}$ D) $\frac{1}{9}$

-
- SR.IIT *CO-SC

72. Area of the trapezium whose vertices lies on the parabola $y^2 = 4x$ and its diagonals pass through $(1,0)$ and having length $\frac{25}{4}$ unit each, is
- A) $\frac{75}{4}$ sq.units B) $\frac{625}{16}$ sq.units C) $\frac{25}{4}$ sq.units D) $\frac{25}{8}$ sq.units
73. Let f be a differentiable function defined in $[0,1]$ such that $f(f(x)) = x$ and $f(0) = 1$. If the value $\int_0^1 (x - f(x))^{2018} dx = \frac{p}{q}$ where p and q are relatively prime then the value of $(p+q)$, is _____
- A) 2023 B) 2024 C) 2020 D) 2022
74. Let $f(x) = \text{Maximum} \{x^2, (1-x)^2, 2x(1-x)\}$, where $0 \leq x \leq 1$. Determine the area of the region bounded by the curves $y = f(x)$, x -axis, $x = 0$ and $x = 1$ is $\frac{A}{54}$ then find the value of A .
- A) 30 B) 36 C) 32 D) 34
75. Let $y = y(x)$ satisfy the differential equation $\left(2xy + x^2y + \frac{y^3}{3}\right)dx + (x^2 + y^2)dy = 0$. If $y(1) = 1$ and the value of $(y(0))^3 = ke$ ($k \in \mathbb{N}$) Then k is _____
- A) 3 B) 4 C) 1 D) 2
76. If $a_1, a_2, \dots, a_{4001}$ are in arithmetic progression and $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{4000} a_{4001}} = 10$ and $a_2 + a_{4000} = 50$. Find the value of $|a_1 - a_{4001}|$.
- A) 20 B) 32 C) 60 D) 30
77. If $a_n = \sum_{r=0}^n \frac{1}{{}^n C_r}$, $b_n = \sum_{r=0}^n \frac{r}{{}^n C_r}$, then the number of ordered pairs (p, q) such that $c_p + c_q = 1$, where $c_p = \frac{a_p}{b_p}$, is:
- A) 0 B) 1 C) 2 D) 3

78. If $(\sin 25^\circ + \sqrt{3} \cos 85^\circ + \sin 85^\circ)^2 = a + b \cos 50^\circ$, a, b , are rational numbers in their simplest form, then $(a + b) =$

- A) 0 B) -2 C) -1 D) 3

79. Let $f(x) = \begin{cases} x+1; & x > 0 \\ 2-x; & x \leq 0 \end{cases}$ and $g(x) = \begin{cases} 3+x; & x < 1 \\ x^2-2x-2; & 1 \leq x < 2 \\ x-5; & x \geq 2 \end{cases}$ then: _____

Note: $[k]$ denotes greatest integer function less than or equal to k .

- A) $\lim_{x \rightarrow 0^+} g(f(x)) = -2$ B) $\lim_{x \rightarrow 0^-} g(f(x)) = -3$
 C) $\lim_{x \rightarrow 0^+} [f(f(x))] = 0$ D) $\lim_{x \rightarrow 0^-} [g(g(x))] = -1$

80. Let f be real-valued function such that $e^{-2x} f(x) = x + 3 + \int_0^x \frac{dt}{\sqrt{t^6 + 1}}$ for all $x \in (-1, 1)$ and let $y = g(x)$ be a function whose graph is reflection of the graph of $y = f(x)$ w. r. t. line $y = x$, then $g'(3)$ is equal to:

- A) 1 B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{1}{8}$

SECTION-II (NUMERICAL VALUE ANSWER TYPE)

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81. Let $A = \begin{bmatrix} x^3+1 & x+5 & 3x+2 \\ y+1 & -6x^2+2 & z-1 \\ 2 & 3 & 9x+6 \end{bmatrix}$ and $|A| = 3$. If $f(x) = \text{tr.}(B^{-1})$ and $B = \text{adj}(A)$, then

global maximum value of $f(x)$ in $x \in [0, 6]$ is:

82. Let $f: A \rightarrow A$ where $A = \{1, 2, 3, 4, 5, 6, 7\}$, then number of function f such that

$f(f(f(x))) = x \forall x \in A$, is:

83. The point on the line $\vec{r} = (-2\vec{i} - 6\vec{j} + 34\vec{k}) + t(2\vec{i} + 3\vec{j} - 10\vec{k}), t \in \mathbb{R}$ which is nearest to the line $\vec{r} = (-6\vec{i} + 7\vec{j} + 7\vec{k}) + \mu(4\vec{i} - 3\vec{j} - 2\vec{k}), \mu \in \mathbb{R}$ is (a, b, c) , then the value of $a + b + c$ is equal to _____
84. Let $I_1 = \int_0^1 \frac{e^x}{1+x} dx$ and $I_2 = \int_0^1 \frac{x^2 dx}{e^{x^3}(2-x^3)}$. Then $\frac{I_1}{e I_2}$ is _____
85. A, B, C are the vertices of triangle with right angled at A and $P(-4, 0); Q(0, 6)$ are two given points. If the ratio of the distances from each vertex of triangle to P, to that of Q is 2:3, the centroid of triangle ABC lies on a circle with radius 'r' then $\left[\frac{r}{\sqrt{13}} \right]$ is equal to _____ [.] represent GIF
86. Let A, B, C, D are four points, in I, II, III, IV quadrants respectively, lying on the circle $(x-1)^2 + (y-2)^2 = 25$ so that AC and BD are perpendicular diameters. Circles, touching the given circle, are drawn at A, B, C, D with radii 1, 2, 3, 4 respectively. If A is (4, 6), then the sum of the coordinates of centers of the circles corresponding to B and D is _____
87. Mr. A either walks to school or take bus to school everyday. The probability that he takes a bus to school is $\frac{1}{4}$. If he takes a bus, the probability that he will be late is $\frac{2}{3}$. If he walks to school, the probability that he will be late is $\frac{1}{3}$. The probability that Mr. A will be on time for at least one out of two consecutive days is $\frac{p}{q}$, where p and q are co-prime, find the value of $(q - p)$.
88. If $\int \frac{\cos e^{2x} - 2019}{(\cos x)^{2019}} dx = \frac{-f(x)}{(g(x))^{2019}} + C$, then find the value of $\left| f\left(\frac{\pi}{4}\right) + g(0) \right|$. (where C is constant of integration)
89. Let $y_1, y_2, y_3, \dots, y_n$ be n observations. Let $w_i = ly_i + k, \forall i = 1, 2, 3, \dots, n$ where l, k are constants. If the mean of y_i 's is 48 and their standard deviation is 12, the mean of w_i 's is 55 and standard deviation of w_i 's is 15, then values of $l + k + 0.75$ should be _____
90. Let f be an invertible function from $\mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation $f^3(x) - (x^3 + 2)f^2(x) + (2x^3 + 1)f(x) - x^3 = 0$. Then the value of $f'(8) \times (f^{-1})'(8)$, is: _____